

**MIDTERM 1 REVIEW SHEET**  
**MATH 411: HONORS COMPLEX VARIABLES, FALL 2026**

**Date:** Friday, October 9, 2026, in class (CAB 369, 3:00–3:50 PM)  
**Length:** 50 minutes  
**Format:** Closed book. No notes, no textbook, no calculators or devices.  
**Coverage:** Lectures 1–12; Lang §§1.1–1.7, 2.1–2.5, 3.2–3.3 (through Goursat)

Since the exam is closed book, you should be able to *state* the main definitions and theorems below precisely, not just recognize them. When you use a result from class in a solution, state clearly which result you are using and check that its hypotheses hold.

Lecture 13 (local primitives, Cauchy’s theorem on a disc) and Lecture 14 (computing real integrals) will *not* be tested on this midterm.

TOPICS

**1. Basic complex numbers** (Lectures 1–3; Lang §§1.1–1.4).

- Arithmetic in  $\mathbb{C}$ : real and imaginary parts, the conjugate  $\bar{z}$ , the modulus  $|z| = \sqrt{z\bar{z}}$ , and the identities  $|zw| = |z||w|$ ,  $\overline{zw} = \bar{z}\bar{w}$ ,  $z^{-1} = \bar{z}/|z|^2$ .
- The triangle inequality  $|z + w| \leq |z| + |w|$  and its reverse form  $||z| - |w|| \leq |z - w|$ .
- Polar form  $z = re^{i\theta}$ ; multiplication multiplies moduli and adds arguments. The definition  $e^{a+ib} = e^a(\cos b + i \sin b)$ .
- Solving  $z^n = w$ : the  $n$  distinct  $n$ -th roots of a nonzero  $w$ , and roots of unity.
- Stereographic projection and the extended plane  $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ .
- Topology of  $\mathbb{C}$ : open and closed discs, open and closed sets, interior, boundary and adherent points; limits and continuity of functions  $f : S \rightarrow \mathbb{C}$  in  $\varepsilon$ – $\delta$  form.

**2.  $\mathbb{C}$ -differentiability** (Lectures 3–5; Lang §§1.5–1.7).

- Definition of  $f'(z_0)$  as a limit over  $h \in \mathbb{C}$ , and of *holomorphic* on an open set. The equivalent “linear approximation” form  $f(z_0 + h) - f(z_0) = f'(z_0)h + h\sigma(h)$  with  $\sigma(h) \rightarrow 0$ .
- Differentiable implies continuous; sum, product, quotient and chain rules. Polynomials are entire; rational functions are holomorphic away from the zeros of the denominator.
- Total (real) differentiability of  $F = (u, v) : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ; the Jacobian; continuous partials  $\Rightarrow$  totally differentiable.
- **Cauchy–Riemann criterion:**  $f = u + iv$  is  $\mathbb{C}$ -differentiable at  $z = x + iy$  if and only if  $F = (u, v)$  is totally differentiable at  $(x, y)$  and  $u_x = v_y$ ,  $u_y = -v_x$  there. In that case  $f' = u_x + iv_x$ . Know the proof, not just the statement.
- Using CR to show a function is (or is not) holomorphic; e.g.  $e^z$  is entire,  $\bar{z}$  and  $|z|^2$  are not holomorphic.
- On a connected open set,  $f' \equiv 0$  implies  $f$  is constant (via “locally constant”). Typical consequences: if  $f$  is holomorphic on a disc and  $\operatorname{Re} f$ ,  $\operatorname{Im} f$  or  $|f|$  is constant, then  $f$  is constant.

### 3. Power series (Lectures 6–9; Lang §§2.1–2.2).

- Formal power series  $\mathbb{C}[[T]]$ : addition, multiplication (Cauchy product), inversion ( $f$  is invertible iff  $a_0 \neq 0$ ), composition  $f \circ g$  when  $g$  has no constant term. Be able to compute the first few coefficients of  $1/f$ ,  $f \cdot g$ ,  $f \circ g$ .
- Convergence of sequences and series in  $\mathbb{C}$ ; Cauchy sequences; absolute convergence; the comparison test.
- Sequences and series of functions: pointwise vs. uniform convergence, the sup norm  $\|\cdot\|_s$ , uniformly Cauchy sequences. A uniform limit of continuous functions is continuous. The comparison test for series of functions (Weierstrass  $M$ -test).
- **Radius of convergence:** every  $\sum a_n z^n$  has  $0 \leq r \leq \infty$  such that it converges absolutely on  $D(0, r)$ , uniformly on each closed disc  $\overline{D}(0, s)$  with  $s < r$ , and diverges for  $|z| > r$ .
- **limsup and Hadamard's formula**  $1/r = \limsup_n |a_n|^{1/n}$ . Be able to compute radii of convergence (including series with gaps, such as  $\sum a_n z^{n^2}$ ), and know the proof.
- Behaviour on the boundary circle can vary: compare  $\sum n z^n$ ,  $\sum z^n/n$ ,  $\sum z^n/n^2$ .
- The formal derivative  $\sum n a_n z^{n-1}$  has the same radius of convergence, and a power series can be differentiated term by term inside its disc of convergence; hence it is infinitely  $\mathbb{C}$ -differentiable there.

### 4. Analytic functions (Lectures 9–11; Lang §§2.3–2.5).

- Definition:  $f$  is *analytic* on  $U$  if near each  $z_0 \in U$  it is given by a convergent power series in  $z - z_0$ .
- A power series with radius  $r$  is analytic on all of  $D(0, r)$  (re-centring a power series).
- Analytic  $\Rightarrow$  holomorphic, in fact infinitely  $\mathbb{C}$ -differentiable; the Taylor coefficients are  $a_k = f^{(k)}(z_0)/k!$ . Sums and products of analytic functions are analytic.
- Uniqueness of power series expansions; **isolated zeros**; the **identity principle** (two convergent power series that agree at a sequence of nonzero points tending to 0 are the same power series).
- Analytic extensions of real functions, and uniqueness of such extensions; the complex exponential, sin, cos.

### 5. Path integrals and the Cauchy–Goursat theorem (Lectures 11–12; Lang §§3.2–3.3).

- Integrals of complex-valued functions of a real variable, and  $|\int_a^b F| \leq \int_a^b |F|$ .
- Piecewise  $C^1$  paths and the path integral  $\int_\gamma f dz = \int_a^b f(\gamma(t)) \gamma'(t) dt$ . Be able to compute integrals over line segments, circles and rectangles (e.g.  $\int_{|z|=1} z^n dz$  for all  $n \in \mathbb{Z}$ ).
- Independence of parametrization; reversing orientation changes the sign.
- **The ML-estimate:**  $|\int_\gamma f| \leq (\sup_\gamma |f|) \cdot L(\gamma)$ .
- For a polynomial  $Q$  and a closed path  $\gamma$ ,  $\int_\gamma Q = 0$ .
- **Cauchy–Goursat:** if  $f$  is holomorphic on an open set  $U$  and  $R \subseteq U$  is a closed rectangle, then  $\int_{\partial R} f = 0$ . Know the structure of the proof: subdivide into four rectangles, pick the worst one, get nested rectangles shrinking to a point  $z_0$ , use the linear approximation of  $f$  at  $z_0$  together with the lemma on polynomials, and finish with the ML-estimate. Note that no continuity of  $f'$  is assumed (compare with the Green's theorem version).